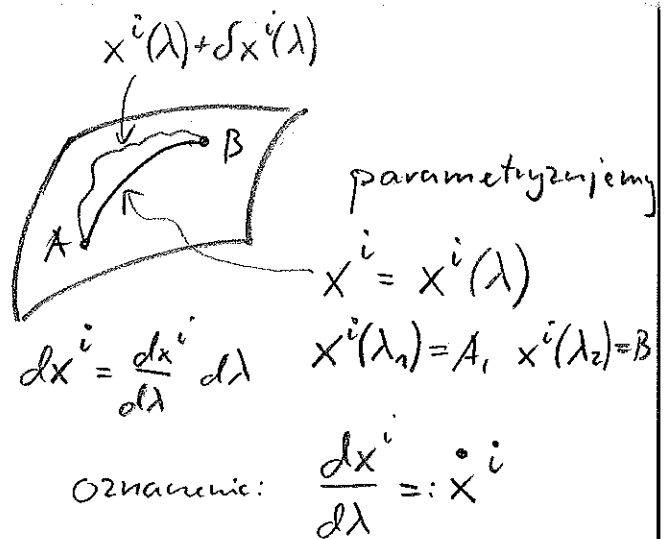


Równanie geodetycznej:

$$0 = \int_A^B \sqrt{g_{ik}(x) dx^i dx^k} = \dots$$

$$\dots = \int_{\lambda_1}^{\lambda_2} \sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k} d\lambda =$$



$$= \int_{\lambda_1}^{\lambda_2} \sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k} d\lambda = \int_{\lambda_1}^{\lambda_2} \frac{\frac{\partial g_{ik}}{\partial x^m} \delta x^m \dot{x}^i \dot{x}^k + g_{ik} \delta \dot{x}^i \dot{x}^k + g_{ik} \dot{x}^i \delta \dot{x}^k}{2 \sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} d\lambda$$

$$= \int_{\lambda_1}^{\lambda_2} \frac{\frac{\partial g_{ik}}{\partial x^m} \delta x^m \dot{x}^i \dot{x}^k + 2 g_{mk} \delta \dot{x}^m \dot{x}^k}{2 \sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} d\lambda = \dots$$

$$\frac{g_{mk} \delta \dot{x}^m \dot{x}^k}{\sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} = \frac{d}{d\lambda} \left(\frac{g_{mk} \delta x^m \dot{x}^k}{\sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} \right) - \delta x^m \frac{d}{d\lambda} \left(\frac{g_{mk} \dot{x}^k}{\sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} \right)$$

$$\dots = \int_{\lambda_1}^{\lambda_2} \delta x^m \left(\frac{1}{2} \frac{\partial g_{ik}}{\partial x^m} \frac{\dot{x}^i \dot{x}^k}{\sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} - \frac{d}{d\lambda} \left(\frac{g_{mk} \dot{x}^k}{\sqrt{g_{ik}(x) \dot{x}^i \dot{x}^k}} \right) \right) d\lambda = \dots$$

parametr λ jest zupełnie dowolny, co powoduje skomplikowaną postać równań L.-E. Tę postać można uprościć wprowadzając jako parametr długości wzdłuż krzywej. Ten wyróżniony parametr można wprowadzić dopiero po obiceniu równań L.-E. z dowolnym parametrem (λ):

$$ds = \sqrt{g_{ik} dx^i dx^k} = \sqrt{g_{ik} \dot{x}^i \dot{x}^k} d\lambda \Rightarrow d\lambda = \frac{ds}{\sqrt{g_{ik} \dot{x}^i \dot{x}^k}}$$

$$\dot{x}^i = \frac{dx^i}{d\lambda} = \frac{dx^i}{ds} \frac{ds}{d\lambda} = \sqrt{g_{ik} \dot{x}^i \dot{x}^k} \frac{dx^i}{ds}$$

$$\dots = \int_{S_1}^{S_2} \delta x^m \frac{1}{2} \frac{\partial g_{ih}}{\partial x^m} \underbrace{\frac{\frac{dx^i}{d\lambda} \frac{dx^h}{d\lambda}}{\sqrt{g_{ih} \dot{x}^i \dot{x}^h}}} \underbrace{\frac{d\lambda}{\sqrt{g_{ih} \dot{x}^i \dot{x}^h}}} \quad -$$

$\frac{dx^i}{ds} \quad \frac{dx^h}{ds}$

$$\int_{S_1}^{S_2} \delta x^m \frac{d}{ds} \left(\frac{g_{mh} \frac{dx^h}{ds}}{\sqrt{g_{ih} \dot{x}^i \dot{x}^h}} \right) ds =$$

$\frac{dx^h}{ds}$

$$= \int_{S_1}^{S_2} \delta x^m \left(\frac{1}{2} \frac{\partial g_{ih}}{\partial x^m} \frac{dx^i}{ds} \frac{dx^h}{ds} - g_{mh} \frac{d^2 x^h}{ds^2} - \frac{\partial g_{mh}}{\partial x^n} \frac{dx^n}{ds} \frac{dx^h}{ds} \right) ds = \dots$$

$\frac{dx^n}{ds} \frac{dx^h}{ds}$ jest symetryczne w indeksach n, h , więc

$$\frac{dx^n}{ds} \frac{dx^h}{ds} \frac{\partial g_{mh}}{\partial x^n} = \frac{1}{2} \frac{dx^n}{ds} \frac{dx^h}{ds} \left(\frac{\partial g_{mh}}{\partial x^n} + \frac{\partial g_{mn}}{\partial x^h} \right)$$

$$\dots = \int_{S_1}^{S_2} \delta x^m (-1) \left(g_{mh} \frac{d^2 x^h}{ds^2} + \frac{1}{2} \left(\frac{\partial g_{mh}}{\partial x^n} + \frac{\partial g_{mn}}{\partial x^h} - \frac{\partial g_{nh}}{\partial x^m} \right) \frac{dx^n}{ds} \frac{dx^h}{ds} \right) ds$$

to musi być 0 aby wariancja $\delta \int_A^B \sqrt{g_{ih} dx^i dx^h} = 0$

otrzymujemy:

$$g_{mh} \frac{d^2 x^h}{ds^2} + \frac{1}{2} \left(\partial_n g_{mh} + \partial_h g_{mn} - \partial_m g_{nh} \right) \frac{dx^n}{ds} \frac{dx^h}{ds} = 0$$

ag

$$\left[\frac{d^2 x^i}{ds^2} + \frac{1}{2} g^{im} \left(\partial_n g_{mh} + \partial_h g_{mn} - \partial_m g_{nh} \right) \frac{dx^n}{ds} \frac{dx^h}{ds} = 0 \right]$$

równanie geodetyczne